TORONTO

REPORT FOR ACTION

Basement Flooding Protection Program Status Update

Date: March 26, 2025
To: Infrastructure and Environment Committee
From: General Manager, Toronto Water
Wards: All

SUMMARY

This report provides a status update on the Basement Flooding Protection Program (BFPP), including the completion of studies and the ongoing and future implementation of BFPP infrastructure upgrades in alignment with the City's recent efforts to enhance capital construction delivery. It also recommends implementing a plan to accelerate project delivery.

Since its inception in 2006, the BFPP has made substantial progress, successfully completing all 67 Basement Flooding studies across the city. These studies have identified nearly 1,200 recommended projects, valued at approximately $18 billion in construction costs, aimed at reducing urban flooding risks. Of those projects identified, approximately 650 were found to meet the program's cost per benefitting property threshold and are eligible to progress to design and construction. At the current rate of project delivery, these projects will take about 50 years to complete. To date, 167 projects have been constructed, reducing flood risk for an estimated 31,000 benefitting properties.

With all studies now complete and following the storm events of the summer of 2024, Council requested an assessment of options to accelerate the construction of the remaining BFPP projects taking into consideration findings from Engineering and Construction Services' review of capital delivery models for large complex municipal infrastructure programs and projects.

The BFPP, delivered by Engineering and Construction Services, will align with the recommendations from the Enhancing the Capital Construction Delivery Process (ECCDP) Initiative, a culmination of industry research, City-led analysis, and Council direction. This may include improvements to project execution through the Integrated Project Management Framework, enhancements to community engagement, refinement of procurement and vendor management, and the exploration of new delivery models and contracts. These measures will increase efficiency and accelerate the delivery of BFPP projects.

Preliminary estimates suggest that if Toronto Water increased annual BFPP construction expenditures by $50 million, and the necessary resources and supports (such as procurement and contract management frameworks) were in place, the delivery rate could increase by 50%. This would benefit an additional 1,000 properties per year and reduce the schedule to complete BFPP projects within the cost per benefitting property threshold from 50 years to approximately 35 years.

RECOMMENDATIONS

The General Manager of Toronto Water recommends that:

1) City Council direct the General Manager, Toronto Water and the Chief Engineer and Executive Director, Engineering and Construction Services, as part of the Enhancing the Capital Construction Delivery Process Initiative, to implement a plan to accelerate delivery of projects identified through the Basement Flooding Protection Program that meet the cost per benefitting property threshold, contingent on the outcome of Recommendation 2.

2) City Council request the General Manager, Toronto Water and the Chief Financial Officer & Treasurer, to report back to City Council through the 2026 Budget Process, on the capital delivery resources and cash flow requirements necessary to support the accelerated delivery of the Basement Flooding Protection Program and highlight any challenges or obstacles constraining acceleration.

FINANCIAL IMPACT

There are no financial impacts from this report.

The 2025-2034 Toronto Water Budget and Capital Plan includes $2.15 billion for the Basement Flooding Protection Program, including studies, engineering, construction, subsidies and complementary programs.

As part of the 2026 budget process, Toronto Water will propose the resource and cash flow requirements to support the accelerated delivery of BFPP projects and will continue to provide a status update of the overall program.

The Chief Financial Officer and Treasurer has reviewed this report and agrees with the financial impact information.

DECISION HISTORY

At its meeting on February 5, 2025, City Council directed Toronto Water and Engineering and Construction Services to report back on delivery and funding options to accelerate the construction of BFPP infrastructure, considering findings from the recent review of capital delivery models, and to undertake operational assessments after storm

events. Item: 2025.EX20.12 - Reducing Stormwater Runoff and Mitigating Basement Flooding

At its meeting on June 5, 2023, the Infrastructure and Environment Committee received a report for information titled “Wet Weather Flow Master Plan Update” which provided an update on the progress of implementing the City's Wet Weather Flow Master Plan, adopted by City Council in 2003, including the progress made in implementing the City's Basement Flooding Protection Program among other programs and initiatives. Item: 2023.IE4.6 - Wet Weather Flow Master Plan Implementation Update

At its meeting on December 6, 2023, the General Government Committee requested Engineering and Construction Services to report back in the fourth quarter of 2024 on its review of capital delivery model options, best practices and recommendations. Item: 2023.GG8.27 Update on Review of Engineering Services Capital Delivery Models Used for Large Complex Municipal Infrastructure Programs and Projects

At its meeting on November 25 and 26, 2020, City Council directed that projects identified through Basement Flooding Studies proceed to detailed design and construction if the cost per benefitting property is less than $68,000, and that projects that exceed the cost per benefitting property threshold be prioritized according to greatest impact criteria. Item: 2020.IE17.5 - Basement Flooding Protection Program Update

At its meeting on March 10 and 11, 2015, City Council directed that Toronto Water initiate and expedite the completion of new Basement Flooding Protection Environmental Assessment (EA) studies for the remainder of the City on a prioritized basis. Item: 2015.EX3.1 - 2015 Rate Supported Budgets - Toronto Water and 2015 Water and Wastewater Rates and Service Fees

The Basement Flooding Protection Program, as a result of the impacts of the July 8, 2013 storm event, moved from being a program based on priority areas to a City-wide program. This change in approach was adopted by City Council at its meeting on December 16, 2013 and the adoption was based on the information contained within a Staff Report titled "Expansion of the Basement Flooding Protection Program's Priority Study Areas". The report can be found at: Expansion of the Basement Flooding Protection Program's Priority Study Areas

At its meeting on September 21 and 22, 2011, City Council adopted criteria for the prioritization of recommended improvement works coming from BFPP studies and preliminary designs, including the criteria of a $32,000 per benefitting property threshold. The staff report noted that the costs of implementing works identified in Environmental Assessments completed among the 32 Chronic Basement Flooding Study Areas far exceed the availability of funding. Item: 2011.PW7.6 - Wet Weather Flow Master Plan and Basement Flooding Protection Program Update

At its meeting on September 24 and 25, 2008, City Council adopted an "Update on the Engineering Review Addressing Basement Flooding", which summarized the results of the engineering analysis, completed to date of the first four of 31 Basement Flooding Study Areas in accordance with the Basement Flooding Work Plan approved by Council

in 2006; and proposed an adaptive management strategy to help reduce the risk of basement flooding, on a City-wide basis. The report also amended the Municipal Code Chapter 681, Sewers to prohibit the connection of downspouts in the basement flooding study areas and all remaining parts of the City. Item: 2008.EX23.16- Update on the Engineering Review Addressing Basement Flooding

At its meeting on April 25 to 27, 2006 City Council adopted a "Work Plan for the Engineering Review Addressing Basement Flooding" focusing on 31 chronic basement flooding areas and established performance criteria for managing drainage due to extreme storm events. Item: Work Plan for the Engineering Review Addressing Basement Flooding (City-wide)

At its meeting on September 22 to 25, 2003, City Council adopted the Wet Weather Flow Management Master Plan and Wet Weather Flow Management Policy. Item: Wet Weather Flow Management Master Plan and Wet Weather Flow Management Policy

COMMENTS

Basement Flooding Protection Program

The Basement Flooding Protection Program (BFPP) is an infrastructure program with the goal of improving the City's resilience to severe storm events by mitigating flooding caused by sewer backups and excessive overland flow. The program is complemented by policies to mitigate basement flooding, public education, and a subsidy program. The BFPP is one of the City's most significant climate adaptation programs.

In 2003, City Council adopted the Wet Weather Flow Master Plan (WWFMP) to reduce and ultimately eliminate the adverse impacts of wet weather flow on Toronto's environment. The WWFMP established 13 objectives including reducing basement flooding, preventing sewer overloading to eliminate/reduce combined sewer overflows, and reducing inflow and infiltration into sanitary sewers.

Following the August 19, 2005 storm, which caused widespread flooding including 3,600 reported basement flooding complaints, erosion in ravines and watercourses, and damage to City infrastructure, City Council adopted a work plan prioritizing areas of chronic basement flooding. This led to 31 Basement Flooding Study Areas. After the July 8, 2013 storm with 4,700 basement flooding complaints, City Council expanded the BFPP City-wide, increasing the number of study areas to 67.

Basement Flooding Studies

Basement Flooding studies assess the capacity of the City's existing storm, sanitary, and combined sewer drainage systems and recommend infrastructure improvements to reduce future basement and surface flooding. These studies follow the Municipal Class Environmental Assessment (EA) standardized planning process as per the Ontario Environmental Assessment Act.

Studies consider a substantial amount of data and engineering analysis to understand conditions and flood risk. Alternative solutions are developed to mitigate flood risk at locations where the BFPP performance criteria are not met. The BFPP applies a Council-adopted performance criteria (program level of service) to assess existing systems:

- Sanitary Sewers: A level of protection against basement flooding caused by sanitary sewer backup based on a design storm equivalent to the May 12, 2000 event as measured at Oriole Yard.
- Municipal Roadways, Combined and Storm Sewers: A level of protection against the City's 100-year storm without excessive ponding or flooding outside of the City's right of way.

Figure 1 presents the results of a computer modelling analysis of Study Area 42, showing where surface ponding is expected during a severe storm event. The model represents the real-world physical environment for the purposes of engineering analysis. Blue coloured triangles represent the accumulation of stormwater during the storm event, with darker shades of blue representing a higher ponding depth.

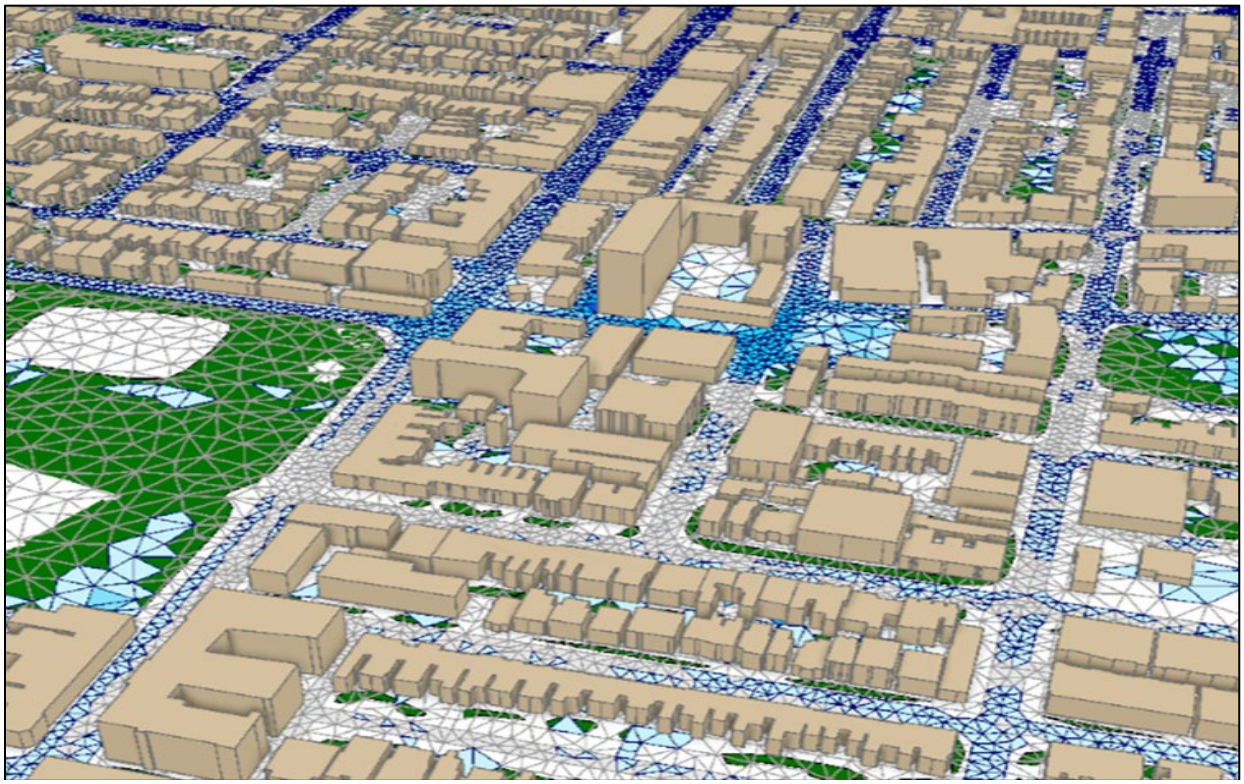


Figure 1 - Overland Flow Analysis - 100-year Storm Event

Figure 2 is a map of the results of a sanitary sewer analysis in Study Area 59. During a severe storm, the red indicates where the sewer level may increase to within basement elevation (1.8 m below the surface of the road), yellow indicates that the sewer is over capacity but basement flooding is not expected, and green indicates the sewer has remaining capacity.

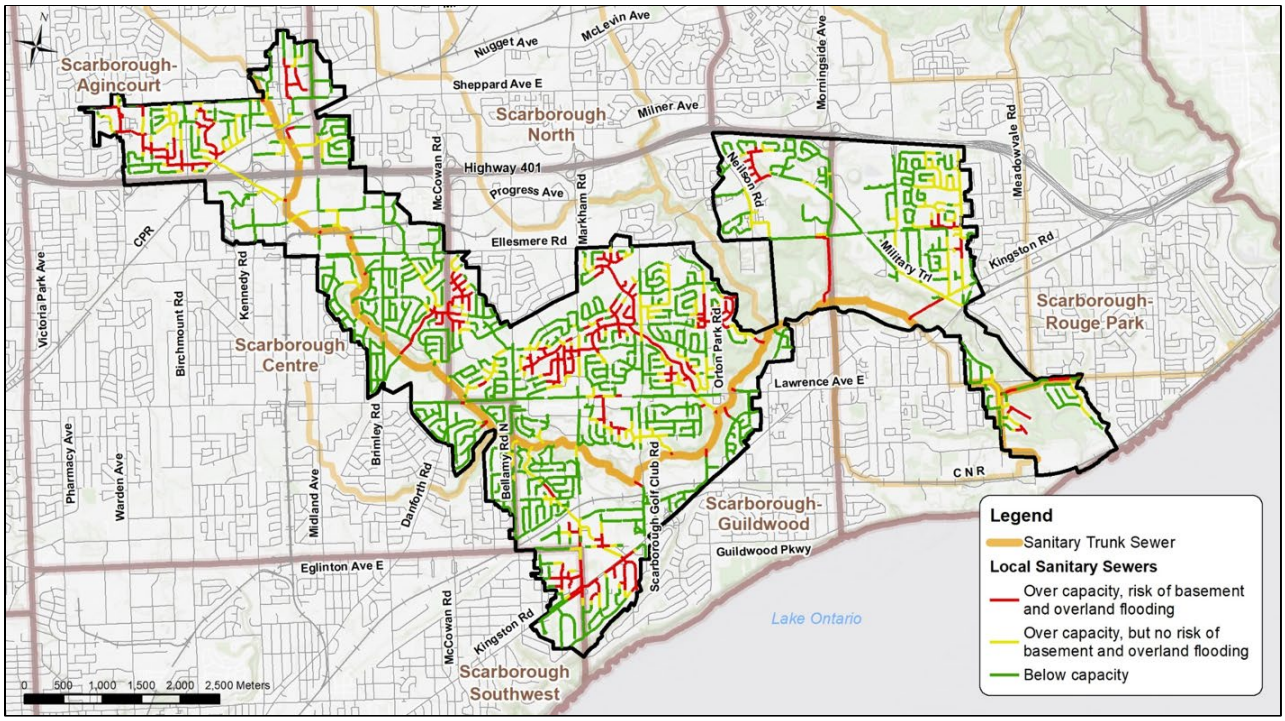


Figure 2 - Sanitary Sewer Analysis - May 12, 2000, event

BFPP proposed solutions may include the construction of larger diameter storm or sanitary sewers, underground and above-ground storage facilities, storm sewer tunnels, catch basins, the removal of inflow and infiltration (I&I)[1], and inlet control devices, which slow the entry of water into the sewer system. Projects may opportunistically include green infrastructure such as bioretention units or oil grit separators to improve water quality. Projects are conceptualized during a Basement Flooding study and refined through preliminary and detailed design.

Riverine flooding is the jurisdiction of the Toronto and Region Conservation Authority. BFPP studies consider its impact on sewer and drainage systems but do not include floodplain mitigation.

Deficiencies in private property drainage systems are not identified and not assessed through the City's Basement Flooding studies. Property owners are responsible for the drainage systems on private property, including lot grading, disconnecting downspouts, private catch basins, private sewer laterals (including tree roots and fats, oils and grease), the operation and maintenance of foundation drains, sump pumps and backwater valves. Toronto Water is currently reviewing new and expanded subsidies under the City’s Basement Flooding Protection Subsidy Program, subject to City Council approval, for, among other things, a stormwater assessment subsidy for private

1 Inflow and infiltration (I&I) is stormwater and groundwater that enters the sanitary sewer system through sump pumps, roof gutter downspouts, foundation drains, and/or cracks in pipes and maintenance holes. I&I can have significant impacts on the capacity of the City's sewer systems and contributes to basement flooding. I&I reduction measures can be beneficial to achieving sewer capacity improvements and reduction of basement flooding occurrences.

property, and will continue to assess future programs targeting private property mitigation measures aimed at reducing factors contributing to basement flooding.

Basement Flooding Study Area Work Plan

In 2006, 31 study areas were identified in chronic flooding areas and a work plan to complete these areas was developed. The BFPP was expanded City-wide, to 67 study areas following the July 8, 2013 storm event and again after the August 7, 2018 storm. The completion of BFPP Studies was accelerated through improved business processes, enabling the City to complete all 67 studies in 2024, ahead of the previously planned completion in 2030.

A map of BFPP study areas is provided in Figure 3, and the completion date of each study is available in Attachment 1. An interactive, web-based map that allows residents to look up their addresses and determine which study area their home is located in can be found at the following link: Basement Flooding Protection Program Map.

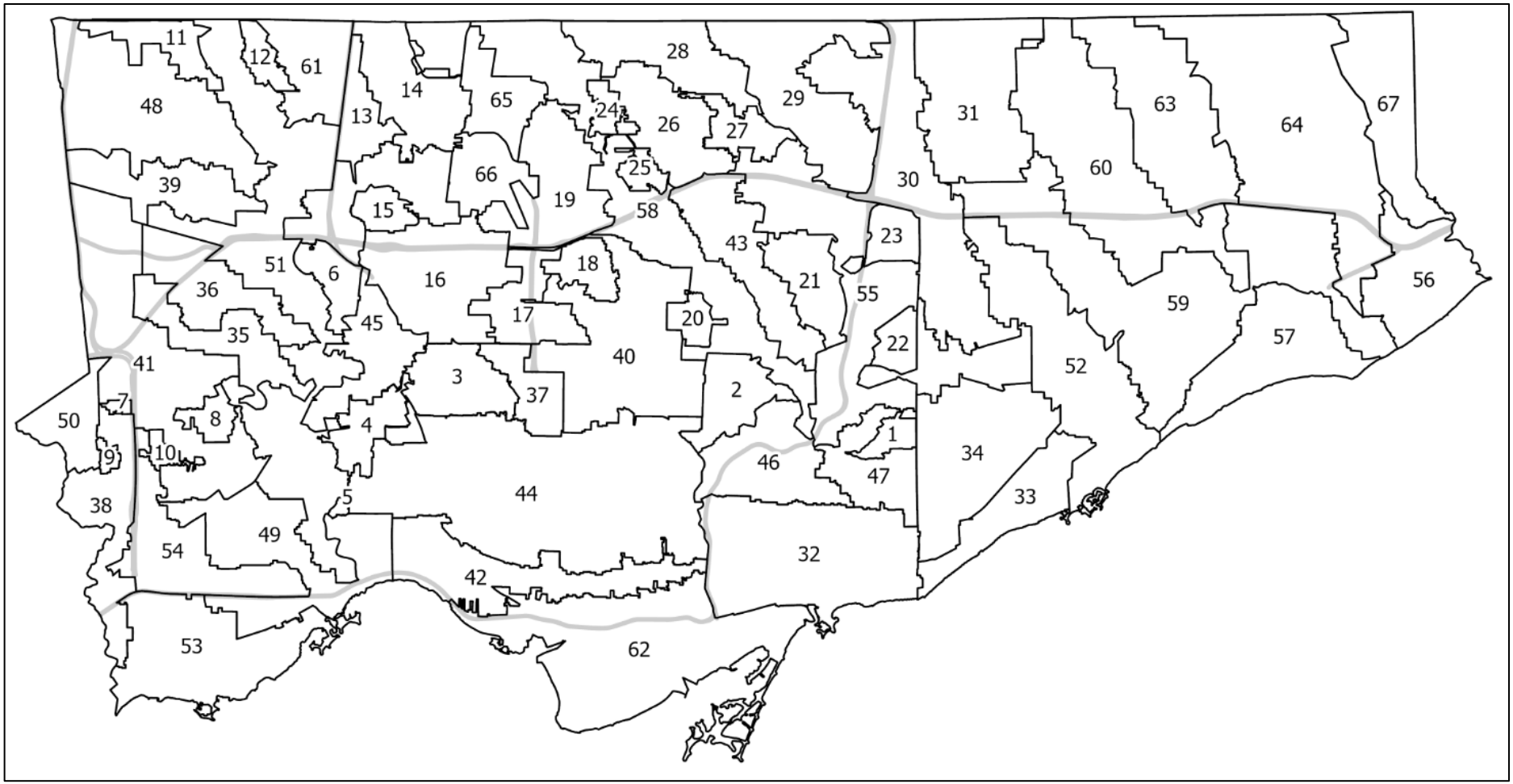


Figure 3 - Map of Basement Flooding Study Areas

A list of the 67 Study Areas and the associated year of study completion is attached in Attachment 1.

Basement Flooding Protection Project Prioritization

Upon completion of the first series of basement flooding studies, the cost to implement complex basement flooding protection projects exceeded available budget. At its meeting on September 24 and 25, 2008, City Council directed that staff prioritize storm drainage projects based on the principle of benefitting the greatest number of properties for expenditure with properties less than $25,000 per benefitting property proceeding through design and construction. In September 2011, City Council directed that the criterion be increased to $32,000 per benefitting property.

On November 25, 2020, City Council directed that the cost per benefitting property criterion be increased to $68,000, and following the completion of all 67 basement flooding studies and once all qualifying projects under $68,000 per benefitting property have been scheduled for implementation, projects will be sequenced based on achieving the greatest impact, considering the following criteria:

- hazard, projects that mitigate a greater extent of flooding and/or a higher probability of flooding;
- exposure, projects that are more likely to have residential basements and/or other critical infrastructure;
- vulnerability, where consideration is given to the Ontario Marginalization Index; and
- project capital cost that considers a lower cost per benefitting property.

Projects that meet the $68,000 cost per benefitting property threshold at the completion of the study stage are eligible to proceed to preliminary design. The preliminary design stage defines the scope and complexity of the work, identifies property and utility impacts, confirms the physical constructability of projects, and refines cost estimates, which may lead to changes in project scope, schedule, and overall costs. Projects still meeting the $68,000 cost per benefitting property threshold after the preliminary design stage proceed to detailed design and construction. Only projects entering the detailed design stage are communicated to the public as projects proceeding to construction.

Projects that exceed the $68,000 cost per benefitting property threshold at the study stage, or at the completion of the preliminary design stage, are added to the deferred projects list. These projects will be prioritized for design and construction in the future based on achieving the greatest impact. Projects may also be sequenced for detailed design and construction as the existing drainage systems warrant replacement due to poor structural condition, or as infrastructure coordination or third-party funding opportunities become available, which materially reduce the City's cost per benefitting property.

Basement Flooding Protection Program - Program Summary

The 67 completed BFPP studies have identified approximately 1,200 recommended projects across the city, with an estimated construction cost of over $18 billion. Figure 4 is a map showing all projects identified through Basement Flooding studies as of January 2025. Project location and scope are subject to change.

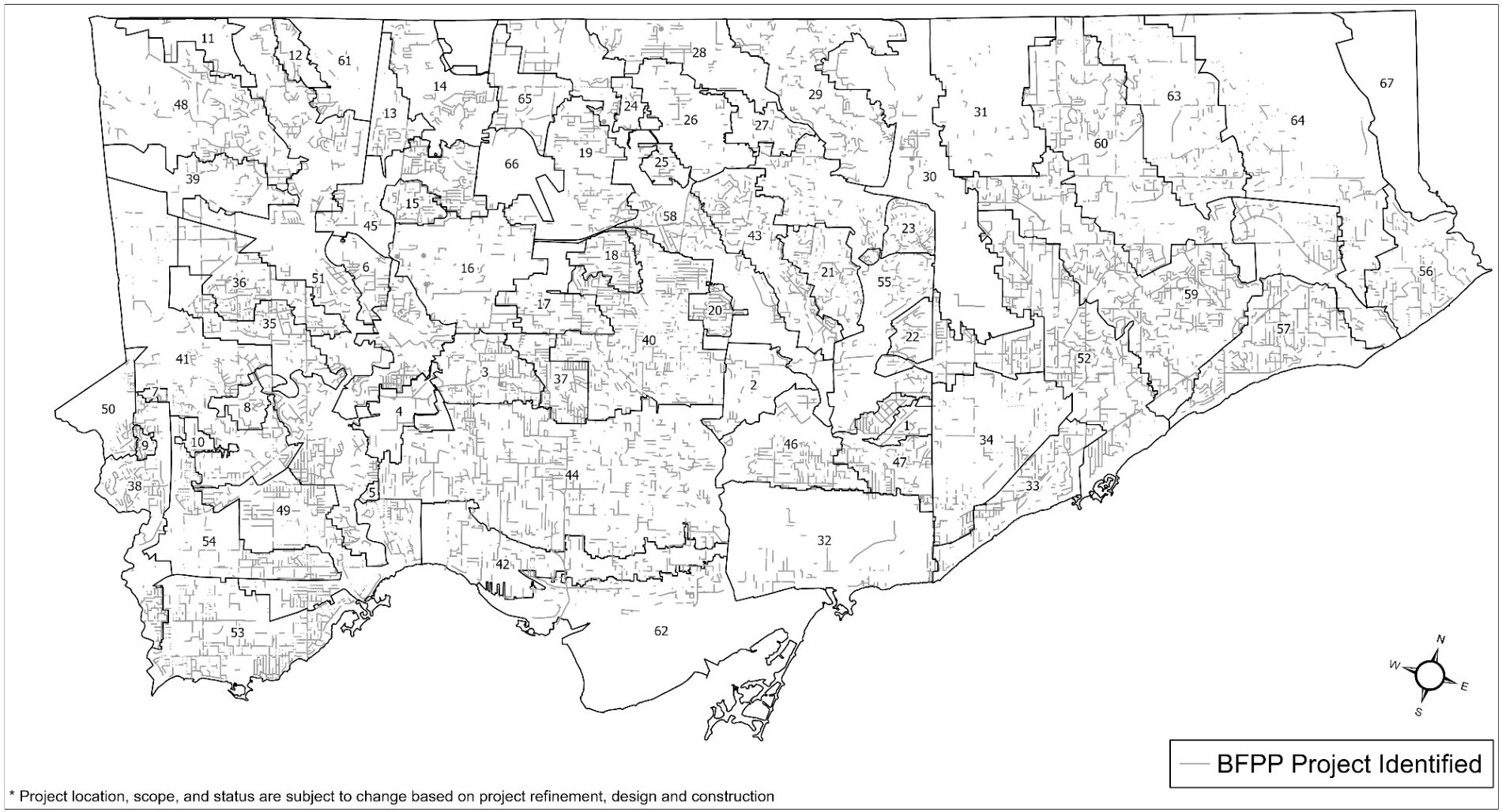


Figure 4 - Basement Flooding Protection Program - Project Summary

As of the end of 2024, approximately $1.15 billion has been spent on 167 BFPP projects. An additional 650 projects, costing an estimated $5 billion, fall below the $68,000 cost per benefitting property. $1.53 billion of construction funding for projects that meet the cost per benefitting property criteria is included in the 2025-2034 Toronto Water Capital Plan.

Approximately 380 projects, exceeding $12 billion, will be scheduled once the projects within the threshold are implemented. It should be noted that project cost estimates are being reviewed and updated, and as such are subject to change.

Since its inception, BFPP construction has mitigated basement flooding risks for more than 31,000 benefitting properties. Ongoing analysis, monitoring and evaluation of completed BFPP projects helps Toronto Water assess the benefit and impact of BFPP projects and mitigation measures.

Following the July 16 and August 17, 2024 storm events, Toronto Water analyzed basement flooding reports in areas where BFPP projects have been constructed. The analysis confirms that projects were successful in reducing flood risks.

For example, in Basement Flooding Study Area 9 (Ward 2, Burnhamthorpe Rd. and Renforth Rd. area), where sanitary and storm sewer replacements and inlet control devices were installed, only one or two basement flooding incidents were reported. The modeling of pre-BFPP conditions predicted significant flooding under comparable storm conditions, but post-upgrade analysis shows the improved infrastructure mitigated this risk. Similarly, in Ward 18, where BFPP sanitary sewer upgrades were completed, only a few isolated incidents were reported.

Basement Flooding Protection Program Implementation

The implementation phase for BFPP projects, which involves design followed by construction, is undertaken by Engineering and Construction Services.

BFPP project photos are provided in Figures 5, 6 and 7.



Figure 5: Left image of a large inline sewer installed in a residential area as part of a BFPP project. Right image of a new sewer installation as part of a BFPP project.



Figure 6: Left image of City officials and contractor in front of the Tunnel Boring Machine for the Fairbank Silverthorn Storm Trunk Sewer Project. Right image of the lowering of the tunnel boring machine into a 40-metre-deep shaft inside Fairbank Memorial Park.



Figure 7: Above image of a dry stormwater management pond with soccer fields in Moore Park, North York (Image sourced from Google Maps, 360 Digilook Inc.). Image below of Old Sheppard underground storage tank.

There are various challenges that impact the completion of BFPP projects at the design and construction phases. These include negotiating easements with property owners, capital coordination with other capital projects in the same area, regulatory approvals, supplier challenges, internal and external technical resource availability, and technical challenges such as design complexities and constructability. In addition, the escalation of costs of the construction of BFPP projects against available budget envelopes has impacted the number of BFPP projects that can be constructed per year.

Accelerating Capital Construction Delivery of the BFPP

Engineering and Construction Services is reporting to Infrastructure and Environment Committee on the Enhancing the Capital Construction Delivery Process (ECCDP) Initiative, consisting of a targeted, research-driven approach informed by Committee and Council motions, a research report from KPMG Consulting with actionable recommendations and implementation plans, and City-led research and analysis. The result is a series of recommendations to improve capital project delivery processes, contract management and procurement processes to better meet the demands of a growing and increasingly complex capital program.

The BFPP will implement recommendations from the Initiative, including improving project execution through the Integrated Project Management Framework, improving community engagement, improving procurement and vendor management, and exploring new delivery models and contracts. The BFPP will also benefit from City efforts to conduct a workforce assessment of the Divisions involved in capital delivery.

In addition to the general, City-wide improvements to capital delivery processes, the BFPP is aiming to increase its rate of capital delivery. On February 5, 2025, City Council directed Toronto Water report to the Infrastructure and Environment Committee on an assessment of delivery and funding options to accelerate the construction of BFPP projects.

At the current rate of expenditure and delivery, it is estimated that construction completion of the remaining projects that meet the cost per benefitting property threshold (costing approximately $5 billion) would take about 50 years. Based on a preliminary estimate, Engineering and Construction Services anticipates that an additional $50 million in annual construction expenditures would result in an estimated 50 per cent increase in construction delivery rate and could equate to more than 1,000 additional properties benefitting per year. This would reduce the construction of completion of BFPP local projects (meeting the cost per benefitting property threshold) from the estimated 50 years to approximately 35 years.

Capital delivery volume is significantly impacted by projects that are delayed or do not proceed to construction. Although many efforts are made through study, conceptualization, design, and construction to mitigate this risk, projects may be impacted by factors such as regulatory approvals, impacts from infrastructure coordination, technical challenges encountered during design and/or construction, and cost escalation resulting in projects being deferred. To reduce the impact on delivery volume, Toronto Water and Engineering and Construction Services will continue to take

steps to introduce a ready project inventory. This will result in additional projects proceeding through design so they can be introduced into the design process should other projects not proceed or be delayed.

As per Council direction, Toronto Water provides an annual update on BFPP progress and a five-year construction plan through the Capital Budget process. With all studies complete, projects meeting the cost per benefitting property criteria will be prioritized into a work plan and schedule.

Aligned with the ECCDP Initiative, BFPP will collaborate with the Strategic Capital Coordination Office to publish a long-term project pipeline ahead of the 2026 capital program. This approach, reflected in the annual BFPP Capital Budget Briefing Note, will enhance industry engagement while acknowledging that long-term project timelines are subject to evolving risks, including scheduling and infrastructure coordination.

CONTACT

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SIGNATURE

Lou Di Gironimo
General Manager, Toronto Water

ATTACHMENTS

Attachment: Basement Flooding Study Area Year of Completion

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Area #	Area Description	Year of Study Completion
1	East York – O'Connor Dr./St. Clair Ave. E.	2014
2	Leaside	2014
3	Keele Street/Eglinton Avenue West	2011
4	Jane St./St. Clair W.	2014
5	Humber River/Bloor St. W.	2014
6	Weston	2014
7	Wellesworth Park	2011
8	Rathburn Rd./Kipling Ave.	2011
9	Burnhamthorpe R. and Renforth Dr. (I&I Study Update in 2017)	2017
10	Glen Park	2011
11	Steeles Ave. W./Martin Grove Rd.	2011
12	Steeles Ave. W./Islington Ave.	2011
13	Glenfield - Jane Heights	2014
14	Black Creek/Finch Ave. West	2009
15	Oakdale - Beverly Heights	2014
16	Black Creek/Highway 401/Dufferin St.	2012
17	Allen Rd. South	2014
18	Highway 401/Bathurst St	2014
19	Clanton Park	2014
20	Lawrence Park	2018
21	Don Mills/York Mills	2016
22	Victoria Village	2014
23	Don Valley Parkway and York Mills	2016
24	Bathurst St/Finch Ave W	2016
25	Lansang - Westgate	2016
26	Willowdale Park	2016
27	Bayview Village	2016
28	Finch Ave. – Bathurst St. to Bayview Ave.	2008
29	West of Hwy 404 – Highway 401 to Steeles Ave. E.	2008
30	Victoria Park Ave.– Eglinton Ave. E. to Steeles Ave. E.	2008
31	Warden Ave. / Finch Ave. E.	2011
32	Eastern Beaches*	2012
33	Scarborough Waterfront (Study Addendum in 2021)	2021
34	Clairlea - Birchmount	2018
35	Silver Creek - Richview - Eglinton Av. W.	2018
36	Kingsview - Humber Creek	2018
37	York – Cedarvale	2019
38	Etobicoke Creek - The West Mall	2018
39	Rexdale Blvd. - The Elms	2018
40	Avenue Rd. – Eglinton Ave. - Yonge St.	2019

Area #	Area Description	Year of Study Completion
41	North Mimico Creek - West Deane - Carlingview Dr.	2018
42	Downtown	2024
43	Bayview and York Mills	2021
44	Mid-Toronto	2024
45	Black Creek	2022
46	East York – Broadview Ave., O'Connor Dr.	2022
47	Taylor-Massey Creek – Wexford	2023
48	Rexdale – Finch Ave. W and Highway 27	2022
49	Islington-Bloor, Park Lawn, South Kingsway	2022
50	Eglinton Ave. W. and Renforth Rd.	2022
51	Royal York Rd./ Kingsway	2023
52	South Scarborough – Highland Creek West	2024
53	Lake Shore Blvd. – Mimico Creek	2023
54	Highway 427 - Queensway	2022
55	Don River – Eglinton	2022
56	Rouge River South	2023
57	South Scarborough – Highland Creek East	2024
58	Keele St./Steeles Ave. to Victoria Park/Eglinton Ave.	2024
59	South Scarborough – Highland Creek	2024
60	Agincourt – Highland Creek West	2023
61	Steeles Ave. W. and Weston Rd.	2022
62	Downtown – Waterfront	2024
63	Agincourt – Highland Creek East	2023
64	Rouge River North	2023
65	Finch Ave. – Allen Rd. to Keele St.	2024
66	Keele St., Sheppard Ave. W., Allen Rd.	2022
67	Steeles Ave. E. and Port Union Rd.	2022

*Area 32 update is underway, targeted completion in early 2026.